

No.	Load condition	Bending moment	Reaction force and shear stress	Deflection
11		$M_C = M_E = \frac{Pl}{4}$ $M_D = \frac{5Pl}{12}$	$R_A = R_B = \frac{3P}{2}$	$d_{max} = \frac{53Pl^3}{1296EI}$
12		$M_C = M_E = \frac{3Pl}{8}$ $M_D = \frac{Pl}{2}$	$R_A = R_B = \frac{3P}{2}$	$d_{max} = \frac{19Pl^3}{348EI}$
13		$M_x = \frac{Wx}{2} (l - \frac{x}{l})$ $M_{max} = \frac{Wl}{8}$	$R_A = R_B = \frac{W}{2}$	$d_{max} = \frac{5}{348} \cdot \frac{Wl^3}{EI}$
14		$M_{max} = \frac{W}{b} (\frac{x_i^2}{2} - a^2)$ where $x_i = a + \frac{Rab}{W}$	$R_A = \frac{W}{l} (\frac{b}{2} + c)$ $R_B = \frac{W}{l} (\frac{b}{2} + a)$	$d_{max} = \frac{W}{384EI} (8l^3 + 4lb^2b^3)$ where $a = c$
15		$M_{max} = \frac{Wa}{4}$	$R_A = R_B = \frac{W}{2}$	$d_{max} = \frac{Wa(3l^3 - 2a^2)}{96EI}$
16		$M_x = \frac{Wx}{3} (l - \frac{x^2}{l^2})$ $M_{max} = 0.128Wl$ where $x_i = 0.5774l$	$R_A = \frac{W}{3}$ $R_B = \frac{2W}{3}$	$d_{max} = \frac{0.01304 Wl^3}{EI}$ where $x = 0.5193l$
17		$M_x = Wx (\frac{l}{2} - \frac{2x^2}{3l})$ $M_{max} = Wl/6$	$R_A = R_B = \frac{W}{2}$	$d_{max} = \frac{Wl^3}{60EI}$
18		$M_{max} = \frac{W}{4} (l - \frac{b}{3})$	$R_A = R_B = \frac{W}{2}$	$d_{max} = \frac{W}{480EI} (8l^3 + 7al^2 + 4a^2l - 4a^3)$
19		$M_{max} = \frac{Wa}{6}$	$R_A = R_B = \frac{W}{2}$	$d_{max} = \frac{Wa}{240EI} (18a^3 + 20ab + 5b^3)$
20		$-M_A = -M_B = M_C = \frac{Pl}{8}$	$R_A = R_B = \frac{P}{2}$	$d_{max} = \frac{Pl^3}{192EI}$

In case of beam subject to distributed loads, load symbol, W, indicates the total load of the beam. For example, incase of a beam subject to uniformly distributed load, w, W equals $w\ell$, where ℓ is span length.